

Section : Biology

Q.1 Arrange the given steps of menstrual cycle in correct sequence.

- (A) Menstruation
- (B) Luteal phase
- (C) Follicular phase
- (D) Next cycle starts

Choose the correct answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (A), (C), (B), (D)
- 3. (B), (A), (D), (C)
- 4. (C), (B), (D), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.2 The technology of biogas production was developed in India mainly due to the efforts of Indian Agricultural Research Institute (IARI) and _____ (KVIC).

- 1. Khadi and Village Industries Corporation
- 2. Khadi and Village Industries Committee
- 3. Khadi and Village Industries Commission
- 4. Khadi and Village Information Commission

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.3 The flower tops, leaves and the resin of cannabis plant are used in various combinations to produce marijuana, hashish, charas and ganja. These are known for their effects on _____ of the body.

- 1. Reproductive system
- 2. Nervous system
- 3. Cardiovascular system
- 4. Digestive system

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.4 Once the BOD of sewage or waste water is reduced significantly, the effluent is then passed into a settling tank where the bacterial 'flocs' are allowed to sediment. This sediment is called as:

- 1. Biogas
- 2. Activated sludge
- 3. Biofertiliser
- 4. Primary sludge

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.5 The transfer of pollen grains from one plant to the stigma of a different plant is known as:

- 1. Geitonogamy
- 2. Xenogamy
- 3. Chasamogamous
- 4. Cleistogamous

Options 1. 1

- 2. 2
- 3. 3

Q.6 Identify the correct sequence involved in the formation of female gametophyte in angiosperms?

- (A) 2-nucleated embryo sac
- (B) Functional megaspore
- (C) Female gametophyte
- (D) 4-nucleated embryo sac

Choose the correct answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (A), (C), (D), (B)
- 3. (B), (A), (D), (C)
- 4. (C), (B), (D), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.7 Arrange the given steps of DNA packaging in correct sequence?

- (A) Histones are organized to form a unit of eight molecules called histone octamer
- (B) Nucleosomes form a structure of repeating unit called chromatin
- (C) A set of positively charged, basic proteins are called histones
- (D) A negatively charged DNA is wrapped around the histone octamer to form a structure known as nucleosome

Choose the correct answer from the options given below:

- 1. (A), (B), (C), (D)
- 2. (A), (C), (B), (D)
- 3. (C), (A), (D), (B)
- 4. (C), (B), (D), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.8 Which of the given parameters contribute to an increase in population density?

- (A) Mortality
- (B) Natality
- (C) Immigration
- (D) Emigration

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (B) and (C) only
- 3. (A), (B) and (C) only
- 4. (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.9 The outer layer of blastocyst is called as:

- 1. Exine
- 2. Trophoblast
- 3. Intine
- 4. Antrum

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.10 Which one of the given diseases has symptoms like stomach pain, constipation, headache, loss of appetite and sustained high fever?

- 1. Malaria
- 2. Pneumonia
- 3. Typhoid

4. Common cold

Options 1. 1

2. 2
3. 3
4. 4

Q.11 Fungi are also known to form symbiotic associations with plants. This is known as-

1. Phycobiont
2. Mycorrhiza
3. Cyanobacteria
4. Antibiotics

Options 1. 1

2. 2
3. 3
4. 4

Q.12 *Trichoderma* species is:

- (A) Free living fungi
- (B) Very common in the root ecosystem
- (C) Effective biocontrol agent
- (D) Resistance to insects

Choose the **correct** answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.13 Which one of the given hormones is secreted from hypothalamus?

1. Insulin
2. TSH
3. GnRH
4. Estrogen

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Match List-I with List-II

List-I

Evolutionary stages of human species

- (A) *Ramapithecus*
- (B) *Australopithecines*
- (C) *Homo erectus*
- (D) Neanderthal man

List-II

Time of origin

- (I) About 1.5 million years ago
- (II) About 15 million years ago
- (III) Between 100,000 - 40,000 years ago
- (IV) Two million years ago

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

- Q.15** Amniotic fluid is used to:
- (A) Determine the survivability of the foetus
 - (B) Massive child immunization
 - (C) Check for pneumonia
 - (D) Check genetic disorder
- Choose the correct answer from the options given below:
- 1. (A) and (D) only
 - 2. (A), (B) and (C) only
 - 3. (A), (B), (C) and (D)
 - 4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

- Q.16** Which drug is an effective painkiller for patients who have undergone surgery?
- 1. Cocaine
 - 2. Opioids
 - 3. Morphine
 - 4. Charas

Options 1. 1

2. 2

3. 3

4. 4

- | Q.17 | List-I | List-II |
|-------------|------------------------------|--|
| | Scientists | Discoveries |
| | (A) Walter Sutton and Boveri | (I) Linkage and recombination of genes on a chromosome |
| | (B) Gregor Mendel | (II) Chromosomal theory of inheritance |
| | (C) Thomas Hunt Morgan | (III) Law of inheritance |
| | (D) Alfred Sturtevant | (IV) Genetic map |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

- Q.18** Match List-I with List-II

Species	Population interaction terms
(A) Monarch butterfly	(I) Brood parasitism
(B) Cuckoo bird	(II) Commensalism
(C) <i>Cuscuta</i>	(III) Camouflage
(D) Sea anemone and Clown fish	(IV) Parasitism

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

3. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

- Q.19** The Hardy - Weinberg Principle says:
- (A) Allele frequency in a population is stable and constant.
 - (B) Allele frequency in a population is unstable.
 - (C) The gene pool remains constant. It is called genetic equilibrium.
 - (D) The sum total of all the Allelic frequencies is 1.
- Choose the correct answer from the options given below:
- 1. (A), (C) and (D) only
 - 2. (A), (B) and (C) only
 - 3. (A), (B), (C) and (D)
 - 4. (B), (C) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

- Q.20** Punnett Square was developed by:
- 1. Gregor Mendel
 - 2. Gregor Punnett
 - 3. Reginald C. Punnett
 - 4. Matthew Meselson Punnett

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

- Q.21** The sex chromosomes of a female bird have:
- 1. XX Chromosomes
 - 2. XY Chromosomes
 - 3. ZW Chromosomes
 - 4. ZZ Chromosomes

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

- Q.22** Zoological parks, botanical gardens and wildlife safari parks are examples of:
- 1. *In situ* conservation
 - 2. *Ex situ* conservation
 - 3. Hotspots
 - 4. Forest conservation

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

- Q.23** Which one of the given structures is not the secondary lymphoid organ?
- 1. Spleen
 - 2. Tonsils
 - 3. Peyer's patches
 - 4. Thymus

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

- Q.24** Identify the correct sequence of given species found in tropical Amazonian rain forest in terms of their increasing number?
- (A) Invertebrates
 - (B) Fishes
 - (C) Birds
 - (D) Plants
- Choose the correct answer from the options given below:
- 1. (A), (B), (C), (D)

2. (A), (D), (C), (B)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.25 Arrange the following trophic levels of the terrestrial ecosystem in correct order?

- (A) Tertiary consumers
- (B) Secondary consumers
- (C) Producers
- (D) Primary consumers

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (D), (B), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.26 The experiment of transforming principle was conducted on:

1. *Streptococcus pneumoniae*
2. Rhino virus
3. *Plasmodium vivax*
4. *Salmonella typhi*

Options 1. 1

2. 2
3. 3
4. 4

Q.27 If in a pond there were 20 lotus plants last year and through reproduction 8 new plants are added, taking the current population to 28, the birth rate is:

1. 1.4 offspring per lotus per year
2. 0.8 offspring per lotus per year
3. 2.5 offspring per lotus per year
4. 0.4 offspring per lotus per year

Options 1. 1

2. 2
3. 3
4. 4

Q.28

List-I

List-II

Enzymes

Products

- | | |
|----------------------------|----------------------------|
| (A) RNA polymerase I | (I) hnRNA |
| (B) RNA polymerase III | (II) hydrolysis of lactose |
| (C) RNA polymerase II | (III) rRNA |
| (D) β -galactosidase | (IV) tRNA, 5SrRNA, snRNA |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3

Q.29 Which one of the following statements is correct about *Saccharomyces cerevisiae*?

1. Used for the production of ethanol
2. Used as a clot buster
3. Used as an immunosuppressive agent
4. Used as blood cholesterol lowering agent

Options 1. 1

2. 2
3. 3
4. 4

Q.30 Which one of them synthesizes and secretes the testicular hormones called androgens?

1. Seminal vesicle
2. Sertoli cells
3. Leydig cells
4. Epididymis

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Identify the correct copper releasing IUDs?

- (A) Cu7
- (B) LNG-20
- (C) CuT
- (D) Multiload 375

Choose the *correct* answer from the options given below:

1. (A), (B) and (D) only
2. (B), (C) and (D) only
3. (A), (B), (C) and (D)
4. (A), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Which one of the following statements does not improve the reproductive health of society?

1. Better detection and cure of STDs
2. Provide post-natal care to increase maternal and infant mortality rates
3. Increase medical facilities for all sex-related problems
4. Increase number of couples with small families

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Which one of the following is not the sacred groves?

1. Khasi and Jaintia Hills in Meghalaya
2. Western Ghat regions of Karnataka and Maharashtra
3. Ajanta and Ellora caves
4. Sarguja, Chanda and Bastar areas of Madhya Pradesh

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Plants can capture only 2-10% of the PAR and this small amount of energy sustains the entire world. Here PAR stands for:

1. Primary active radiation
2. Photosynthetically active reflection

3. Photosynthetically active radiation
4. Photosynthetically active refraction

Options 1. 1

2. 2
3. 3
4. 4

Q.35 The unequivocal proof that DNA is the genetic material came from the experiments of -

1. Frederick Griffith
2. Alfred Hershey and Martha Chase
3. Matthew Meselson and Franklin Stahl
4. Maurice Wilkins and Rosalind Franklin

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Which one of the given approaches is used when the animals and plants are endangered?

1. *In situ conservation*
2. *Ex situ conservation*
3. Endemism
4. Alien species introduction

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Jawless fish probably evolved around:

1. 320 million years ago
2. 350 million years ago
3. 500 million years ago
4. 700 million years ago

Options 1. 1

2. 2
3. 3
4. 4

Q.38 Which one of the following seeds is an example of non-albuminous seed?

1. Wheat
2. Barley
3. Peas
4. Castor

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Common cold is caused by:

1. *Streptococcus pneumoniae*
2. Rhino virus
3. *Haemophilus influenzae*
4. *Microsporum*

Options 1. 1

2. 2
3. 3
4. 4

Q.40 Match List-I with List-II

List-I

List-II

Terms	Features
(A) Humus	(I) Mass of living organisms
(B) Detritus	(II) Dark-colored amorphous substance
(C) Producers	(III) Dead remains of plants and animals
(D) Biomass	(IV) Phytoplankton

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the following paragraph and answer the given questions.

Repeated amplification is achieved by the use of a thermostable DNA polymerase (isolated from a bacterium, *Thermus aquaticus*), which remain active during the high temperature induced denaturation of double stranded DNA. The separated DNA fragments can be visualised only after staining the DNA with a compound known as ethidium bromide followed by exposure to UV radiation. The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. The convention for naming these enzymes is the first letter of the name comes from the genus and the second two letters come from the species of the prokaryotic cell from which they were isolated, e.g., EcoRI comes from *Escherichia coli* RY 13. In EcoRI, the letter 'R' is derived from the name of strain. Roman numbers following the names indicate the order in which the enzymes were isolated from that strain of bacteria. Stanley Cohen and Herbert Boyer accomplished this in 1972 by isolating the antibiotic resistance gene by cutting out a piece of DNA from a plasmid which was responsible for conferring antibiotic resistance. Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA, RNA, protein, polysaccharides etc. The RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol.

In agarose gel electrophoresis, the separated DNA fragments can be visualized with the help of compound known as:

1. Ethidium chloride
2. Rubidium bromide
3. Ethidium bromide
4. Caesium bromide

Options 1. 1

2. 2
3. 3
4. 4

Q.42 Read the following paragraph and answer the given questions.

Repeated amplification is achieved by the use of a thermostable DNA polymerase (isolated from a bacterium, *Thermus aquaticus*), which remain active during the high temperature induced denaturation of double stranded DNA. The separated DNA fragments can be visualised only after staining the DNA with a compound known as ethidium bromide followed by exposure to UV radiation. The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. The convention for naming these enzymes is the first letter of the name comes from the genus and the second two letters come from the species of the prokaryotic cell from which they were isolated, e.g., EcoRI comes from *Escherichia coli* RY 13. In EcoRI, the letter 'R' is derived from the name of strain. Roman numbers following the names indicate the order in which the enzymes were isolated from that strain of bacteria. Stanley Cohen and Herbert Boyer accomplished this in 1972 by isolating the antibiotic resistance gene by cutting out a piece of DNA from a plasmid which was responsible for conferring antibiotic resistance. Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA, RNA, protein, polysaccharides etc. The RNA can be removed by treatment with

ribonuclease whereas proteins can be removed by treatment with protease. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol.

In the PCR, the thermostable DNA polymerase enzyme is obtained from:

1. *Thermus acuaticus*
2. *Thermus aquaticus*
3. *Thermus acquaticus*
4. *Thermus aquaticas*

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Read the following paragraph and answer the given questions.

Repeated amplification is achieved by the use of a thermostable DNA polymerase (isolated from a bacterium, *Thermus aquaticus*), which remain active during the high temperature induced denaturation of double stranded DNA. The separated DNA fragments can be visualised only after staining the DNA with a compound known as ethidium bromide followed by exposure to UV radiation. The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. The convention for naming these enzymes is the first letter of the name comes from the genus and the second two letters come from the species of the prokaryotic cell from which they were isolated, e.g., EcoRI comes from *Escherichia coli* RY 13. In EcoRI, the letter 'R' is derived from the name of strain. Roman numbers following the names indicate the order in which the enzymes were isolated from that strain of bacteria. Stanley Cohen and Herbert Boyer accomplished this in 1972 by isolating the antibiotic resistance gene by cutting out a piece of DNA from a plasmid which was responsible for conferring antibiotic resistance. Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA, RNA, protein, polysaccharides etc. The RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol.

Stanley Cohen and Herbert Boyer constructed the first recombinant DNA by isolating the antibiotic resistance gene by cutting out a piece of DNA from a plasmid of _____ resistance. which was responsible for conferring antibiotic

1. *Thermus aquaticus*
2. *Agrobacterium tumifaciens*
3. *Escherichia coli*
4. *Salmonella typhimurium*

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Read the following paragraph and answer the given questions.

Repeated amplification is achieved by the use of a thermostable DNA polymerase (isolated from a bacterium, *Thermus aquaticus*), which remain active during the high temperature induced denaturation of double stranded DNA. The separated DNA fragments can be visualised only after staining the DNA with a compound known as ethidium bromide followed by exposure to UV radiation. The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. The convention for naming these enzymes is the first letter of the name comes from the genus and the second two letters come from the species of the prokaryotic cell from which they were isolated, e.g., EcoRI comes from *Escherichia coli* RY 13. In EcoRI, the letter 'R' is derived from the name of strain. Roman numbers following the names indicate the order in which the enzymes were isolated from that strain of bacteria. Stanley Cohen and Herbert Boyer accomplished this in 1972 by isolating the antibiotic resistance gene by cutting out a piece of DNA from a plasmid which was responsible for conferring antibiotic resistance. Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA, RNA, protein, polysaccharides etc. The RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease.

Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol.

The convention for naming restriction enzymes is:

1. First three letter of the name from the genus and second two letters from the species of bacterium from which they were isolated
2. First two letter of the name from the species and second two letters from the genus of bacterium from which they were isolated
3. First letter of the name from the species and second two letters from the genus of bacterium from which they were isolated
4. First letter of the name from the genus and second two letters from the species of bacterium from which they were isolated

Options 1. 1

2. 2

3. 3

4. 4

Q.45 Read the following paragraph and answer the given questions.

Repeated amplification is achieved by the use of a thermostable DNA polymerase (isolated from a bacterium, *Thermus aquaticus*), which remain active during the high temperature induced denaturation of double stranded DNA. The separated DNA fragments can be visualised only after staining the DNA with a compound known as ethidium bromide followed by exposure to UV radiation. The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. The convention for naming these enzymes is the first letter of the name comes from the genus and the second two letters come from the species of the prokaryotic cell from which they were isolated, e.g., EcoRI comes from *Escherichia coli* RY 13. In EcoRI, the letter 'R' is derived from the name of strain. Roman numbers following the names indicate the order in which the enzymes were isolated from that strain of bacteria. Stanley Cohen and Herbert Boyer accomplished this in 1972 by isolating the antibiotic resistance gene by cutting out a piece of DNA from a plasmid which was responsible for conferring antibiotic resistance. Since the DNA is enclosed within the membranes, we have to break the cell open to release DNA, RNA, protein, polysaccharides etc. The RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease. Other molecules can be removed by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol.

In the isolation process of genetic material, the purified DNA can be precipitated out after the addition of:

1. Chilled water
2. Chilled methanol
3. Chilled ethanol
4. Warm ethanol

Options 1. 1

2. 2

3. 3

4. 4

Q.46 Read the following paragraph and answer the given questions.

RNAi takes place in all eukaryotic organisms as a method of cellular defense. The source of this complementary RNA could be from an infection by viruses having RNA genomes or mobile genetic elements (transposons) that replicate via an RNA intermediate. Insulin from an animal source, though caused some patients to develop allergy or other types of reactions to the foreign protein. Insulin consists of two short polypeptide chains: chain A and chain B, that are linked together by disulphide bridges. Transgenic mice are being developed for use in testing the safety of vaccines before they are used on humans. Transgenic mice are being used to test the safety of the polio vaccine. If successful and found to be reliable, they could replace the use of monkeys to test the safety of batches of the vaccine. Specific Bt toxin genes were isolated from *Bacillus thuringiensis* and incorporated into the several crop plants such as cotton. The choice of genes depends upon the crop and the targeted pest, as most Bt toxins are insect-group specific. The toxin is coded by a gene *cryIAc* named cry. There are a number of them, for example, the proteins encoded by the genes *cryIAc* and *cryIIAb* control the cotton bollworms, that of *cryIAb* controls corn borer.

The suitable example of mobile genetic elements is:

1. dsRNA
2. Vector
3. Transposons
4. Plasmids

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the following paragraph and answer the given questions.

RNAi takes place in all eukaryotic organisms as a method of cellular defense. The source of this complementary RNA could be from an infection by viruses having RNA genomes or mobile genetic elements (transposons) that replicate via an RNA intermediate. Insulin from an animal source, though caused some patients to develop allergy or other types of reactions to the foreign protein. Insulin consists of two short polypeptide chains: chain A and chain B, that are linked together by disulphide bridges. Transgenic mice are being developed for use in testing the safety of vaccines before they are used on humans. Transgenic mice are being used to test the safety of the polio vaccine. If successful and found to be reliable, they could replace the use of monkeys to test the safety of batches of the vaccine. Specific Bt toxin genes were isolated from *Bacillus thuringiensis* and incorporated into the several crop plants such as cotton. The choice of genes depends upon the crop and the targeted pest, as most Bt toxins are insect-group specific. The toxin is coded by a gene *cryIAc* named cry. There are a number of them, for example, the proteins encoded by the genes *cryIAc* and *cryIIAb* control the cotton bollworms, that of *cryIAb* controls corn borer.

The transgenic mice are being used to test the safety of which vaccine?

1. Polio vaccine
2. Hepatitis vaccine
3. Measles vaccine
4. Rabies vaccine

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the following paragraph and answer the given questions.

RNAi takes place in all eukaryotic organisms as a method of cellular defense. The source of this complementary RNA could be from an infection by viruses having RNA genomes or mobile genetic elements (transposons) that replicate via an RNA intermediate. Insulin from an animal source, though caused some patients to develop allergy or other types of reactions to the foreign protein. Insulin consists of two short polypeptide chains: chain A and chain B, that are linked together by disulphide bridges. Transgenic mice are being developed for use in testing the safety of vaccines before they are used on humans. Transgenic mice are being used to test the safety of the polio vaccine. If successful and found to be reliable, they could replace the use of monkeys to test the safety of batches of the vaccine. Specific Bt toxin genes were isolated from *Bacillus thuringiensis* and incorporated into the several crop plants such as cotton. The choice of genes depends upon the crop and the targeted pest, as most Bt toxins are insect-group specific. The toxin is coded by a gene *cryIAc* named cry. There are a number of them, for example, the proteins encoded by the genes *cryIAc* and *cryIIAb* control the cotton bollworms, that of *cryIAb* controls corn borer.

Identify the incorrect fact about the genetic modification?

1. It has made crops more tolerant to abiotic stresses (cold, drought, salt, heat).
2. It has helped to reduce post harvest losses.
3. It has decreased the nutritional value of food, e.g., golden rice, i.e., Vitamin 'A' enriched rice.
4. It has reduced reliance on chemical pesticides (pest-resistant crops).

Options 1. 1

2. 2
3. 3

Q.49 Read the following paragraph and answer the given questions.

RNAi takes place in all eukaryotic organisms as a method of cellular defense. The source of this complementary RNA could be from an infection by viruses having RNA genomes or mobile genetic elements (transposons) that replicate via an RNA intermediate. Insulin from an animal source, though caused some patients to develop allergy or other types of reactions to the foreign protein. Insulin consists of two short polypeptide chains: chain A and chain B, that are linked together by disulphide bridges. Transgenic mice are being developed for use in testing the safety of vaccines before they are used on humans. Transgenic mice are being used to test the safety of the polio vaccine. If successful and found to be reliable, they could replace the use of monkeys to test the safety of batches of the vaccine. Specific Bt toxin genes were isolated from *Bacillus thuringiensis* and incorporated into the several crop plants such as cotton. The choice of genes depends upon the crop and the targeted pest, as most Bt toxins are insect-group specific. The toxin is coded by a gene *cryIAc* named cry. There are a number of them, for example, the proteins encoded by the genes *cryIAc* and *cryIIAb* control the cotton bollworms, that of *cryIAb* controls corn borer.

Which of these is not present in mature insulin?

1. Disulfide bonds
2. C peptide
3. A peptide
4. B peptide

Options 1. 1

2. 2

3. 3

4. 4

Q.50 Read the following paragraph and answer the given questions.

RNAi takes place in all eukaryotic organisms as a method of cellular defense. The source of this complementary RNA could be from an infection by viruses having RNA genomes or mobile genetic elements (transposons) that replicate via an RNA intermediate. Insulin from an animal source, though caused some patients to develop allergy or other types of reactions to the foreign protein. Insulin consists of two short polypeptide chains: chain A and chain B, that are linked together by disulphide bridges. Transgenic mice are being developed for use in testing the safety of vaccines before they are used on humans. Transgenic mice are being used to test the safety of the polio vaccine. If successful and found to be reliable, they could replace the use of monkeys to test the safety of batches of the vaccine. Specific Bt toxin genes were isolated from *Bacillus thuringiensis* and incorporated into the several crop plants such as cotton. The choice of genes depends upon the crop and the targeted pest, as most Bt toxins are insect-group specific. The toxin is coded by a gene *cryIAc* named cry. There are a number of them, for example, the proteins encoded by the genes *cryIAc* and *cryIIAb* control the cotton bollworms, that of *cryIAb* controls corn borer.

The Bt toxin gene *cryIAb* controls the:

1. Cotton bollworm
2. Corn borer
3. Tobacco budworm
4. Armyworm

Options 1. 1

2. 2

3. 3

4. 4